

DHRUV ARORA

Software Developer 2

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PROFILE

Full Stack Developer with 3+ years of experience building scalable backend systems and performant frontend applications. Promoted to SDE-II within 3 years, with a track record of owning complex problems end-to-end - from low-level native system integrations to real-time collaborative platforms.

TECHNICAL SKILLS

Backend: Java, Spring Boot, PostgreSQL, Elasticsearch, Redis, WebSockets

Frontend: TypeScript, React.js, ElectronJS, Vanilla JavaScript

Systems: C++, Objective-C, Node Addon API (NAPI)

Libraries & Tools: YJS (CRDT), Excalidraw, draw.io, Monaco Editor, MessagePack

Domains: Real-time Collaboration, Secure Browser Engineering, Microservices

PROFESSIONAL EXPERIENCE

Software Developer II | *Mercer Mettl*

July 2022 – Present

Working as a full stack developer contributing across the entire stack from backend services to frontend interfaces. Actively participates in system design, architecture discussions, mentoring new developers.

Mettl Secure Browser - MacOS

- Leading end-to-end development alongside a junior developer, owning architecture, implementation, and delivery.
- Core security features built as native Node.js addons using NAPI in C++ and Objective-C, enabling deep macOS system integration within the ElectronJS shell.
- Implemented observer/notification-based event handling instead of polling, reducing CPU overhead with near-instant system event response.
- Security controls include: screen capture/share detection, VM detection, overlay prevention, app blacklisting/whitelisting, and keystroke monitoring to block unauthorised shortcuts.

Interviews Platform

- Consolidated two Redis Pub/Sub microservices into one unified messaging server, reducing infrastructure costs and simplifying scaling - used for backend notifications and real-time peer-to-peer collaboration.
- Designed a custom binary WebSocket protocol using bit-flag headers and length-prefixed fields for compact, efficient messaging; JSON payloads are further compressed via MessagePack. Shipped as an npm package (frontend) and Java module (backend), each with a custom serialiser/deserialiser.
- Built a suite of real-time collaborative editors using custom YJS adapters - Whiteboard (Excalidraw), Diagram Tool (draw.io), and Coding Tool (Monaco Editor) - each implemented as pluggable React components in a dedicated UI library, with service-level logic abstracted into a separate TypeScript library.
- Improved frontend performance through bundle size reduction and code splitting, cutting initial Time to First Display (TTFD).

Senior Analyst (Intern) | *Capgemini*

January 2022 – May 2022

- Worked with SAP Process Integration/Process Orchestration (PI/PO) and SAP Cloud Platform Integration (CPI) to design and implement data integration workflows.

PUBLICATIONS

Machine Learning-Assisted Binary and Multi-Class Parkinson's Disease Detection

Presented at ICISS 2022

Developed machine learning models to enhance clinical detection of Parkinson's disease across binary and multi-class datasets.

EDUCATION

B.Tech in Computer Science | *Manipal University Jaipur*

2018 – 2022

CGPA: 9.2 / 10

High School (XII) | *Delhi Public School, Faridabad*

2018